Steps in Logarithmic Differentiation

(1) Take natural logarithm on both sides of an equation $y = f(x)$ and use the law of logarithms to simplify.

(2) Differentiate implicitly with respect to x.

(3) Solve the resulting equation for y'.

In general there are four cases for exponents and bases.

(1) $\frac{d}{dx}(a^b) = 0$ (a and b are constants).

(2) $\frac{d}{dx}[f(x)]^b = b[f(x)]^{b-1} f'(x)$

(3) $\frac{d}{dx}[a^{g(x)}] = a^{g(x)}(\log a)g'(x)$

(4) $\frac{d}{dx}[f(x)]^{g(x)} = [f(x)]^{g(x)}\left[\frac{g(x)f'(x)}{f(x)} + \log f(x).g'(x)\right]$

Example 10.23

Differentiate $y = x^{\sqrt{x}}$.

Solution

Take logarithm :

$$\log y = \sqrt{x}\log x$$

Differentiating implicitly,

$$\frac{y'}{y} = \sqrt{x}.\frac{1}{x} + \frac{1}{2\sqrt{x}}.\log x$$

$$= \frac{\log x + 2}{2\sqrt{x}}$$

Therefore, $\frac{d}{dx}(x^{\sqrt{x}}) = y' = x^{\sqrt{x}}\left(\frac{\log x + 2}{2\sqrt{x}}\right)$.

10.4.5 Substitution method

It is very much useful in some processes of differentiation, in particular the differentiation involving inverse trigonometrical functions.

Consider $f(x) = \tan^{-1}\left(\frac{2x}{1-x^2}\right)$.

For this function $f'(x)$ can be found out by using function of a function rule. But it is laborious. Instead we can use the substitution method.

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